PROJECT ONE - Analysis of Student Success Data

BACKGROUND AND OVERVIEW

This project investigates the relationship between psychological stressors and academic outcomes among high school students. Using SQL to merge and analyze disparate datasets, the study aims to identify at-risk student populations and evaluate the effectiveness of institutional support systems.

DATA STRUCTURE AND OVERVIEW

The analysis was performed in Google BigQuery using a relational schema consisting of two primary datasets:

student_performance_finalscore	
Student_ID	STRING
Age	INT64
Gender	STRING
Hours_Studied	NUMERIC
Attendance	NUMERIC
Sleep_Hours	NUMERIC
Stress_Level	NUMERIC
Screen_Time	NUMERIC
Previous_GPA	NUMERIC
Part_TIme_Job	STRING
Study_Method	STRING
Diet_Quality	STRING
Internet_Quality	STRING
Extracurricular	STRING
Tutoring_Sessions_Per_Week	INT64
Family_Income	STRING
Exam_Anxiety_Score	NUMERIC
Final_Score	STRING

student_performance	
Student_ID	STRING
Age	INT64
Gender	STRING
Hours_Studied	NUMERIC
Attendance	NUMERIC
Sleep_Hours	NUMERIC
Stress_Level	NUMERIC
Screen_Time	NUMERIC
Previous_GPA	NUMERIC
Part_TIme_Job	STRING
Study_Method	STRING
Diet_Quality	STRING
Internet_Quality	STRING
Extracurricular	STRING
Tutoring_Sessions_Per_Week	INT64
Family_Income	STRING
Exam_Anxiety_Score	NUMERIC
Grade	STRING

Key Technical Operation: Tables were joined via a Student_ID key.

EXECUTIVE SUMMARY

The data reveals that academic success is a multi-variable equation where both psychological health and support systems are as predictive as students' natural academic aptitude. This project identifies 'Threshold Anxiety' at the C/D GPA border as a primary predictor of academic failure and utilizes CTEs to evaluate institutional safeguards and propose targeted, data-driven support strategies.

- A moderate negative correlation ($r=-0.49$) exists between exam anxiety and final scores.
- Anxiety levels account for 24% of the variance in final grades ($r2=0.24$), suggesting that nearly one-quarter of a student's performance is dictated by their stress state.
- Students exhibit the highest levels of anxiety (13% higher exam anxiety scores than compared to the “safe group”) when in proximity to the “C/D” GPA border.
- Students who received formal support achieved an average final score of 85.51, representing a 20.55% increase over the non-tutored group average of 70.93.
- There is a non-linear relationship between sleep and anxiety and student performance. Within “C/D” borderline students, “Poorly Rested” students out performed their “Well Rested” counterparts but also exhibited significantly higher anxiety scores.

INSIGHTS DEEP DIVE

The C/D Threshold Spike

Data reveals a significant non-linear spike in stress as students approach the passing threshold. Students in the "Aspiring D" category (those fighting to move from academic probation to safety) reported anxiety levels 18.5% higher than those in the "Safe" category.

- Insight: The psychological pressure is highest at the point of "Academic Survival." Interestingly, anxiety levels significantly decrease once students move into the "Aspiring C" tier (4.67), suggesting that once the immediate threat of failing is mitigated, the intense "survival stress" begins to subside.
- Emotional Volatility and the Breaking Point: While anxiety peaks at the bottom of the scale, the Anxiety Sigma identifies where emotional states are most unpredictable. The "Aspiring D" group shows the highest volatility (Sigma 1.87), suggesting a subset of students reaching a breaking point. Conversely, "Honors" tracks (Aspiring/At Risk) show the most uniform stress responses (Sigma ~1.60), identifying them as having more predictable coping mechanisms despite their high-stakes environment.

- Conclusion: For students at the 1.9–2.1 GPA threshold, academic struggle is a primary catalyst for emotional instability. However, for the majority, distance from the "C/D" borderline results in the most significant psychological gain: a measurable drop in anxiety and a more stable emotional profile. Intervention strategies should prioritize students in closest proximity to the “C/D” borderline to maximize both performance and student wellness.

Tutoring as a Stabilizer

The data indicates that institutional intervention (tutoring) provides a dual benefit: it raises the academic performance while simultaneously stabilizing the emotional health of students. Across the total population, tutoring acts as a vital safeguard against performance volatility.

- Performance & Volatility Reducer: Students who utilized tutoring achieved an average final score of 85.51, a 20.55% increase over the non-tutored average of 70.93. Beyond higher scores, tutoring significantly tightened performance; the Score Sigma for tutored students was 10.91, compared to 14.72 for those without support. This represents a 25.88% reduction in performance variance, making academic outcomes far more predictable.
- Anxiety Reduction: Tutoring correlates with a 17.66% reduction in average anxiety scores. Furthermore, non-tutored students exhibited significantly more emotional volatility (Sigma 1.78) compared to their tutored counterparts (Sigma 1.62). This stabilization is most dramatic in the C/D Borderline group, where tutoring reduced peak anxiety by 23.69%, successfully pulling high-risk students back from a psychological breaking point.
- Conclusion: Tutoring acts as a primary stabilizer. By reducing performance volatility by nearly 26%, tutored students are significantly less likely to experience "outlier" failing grades. It transforms the student experience from a high-stakes struggle into a consistent, manageable academic trajectory.

Lifestyle Safeguards

In the C/D Borderline group, the relationship between sleep and performance shifts significantly. Poorly Rested students achieved the highest average scores (64.91), yet reported the highest anxiety levels (6.77). This indicates an effect where students sacrifice emotional well-being and rest to achieve the marginal gains necessary for academic survival.

- Volatility and the Exhaustion Gap: While the Poorly Rested group scored higher, they exhibited extreme performance instability with a Score Sigma of 20.72—the highest in the study. In contrast, the Moderately Rested group showed much greater consistency (Sigma 13.25). This suggests that while sleep deprivation can lead to short-term score spikes, it creates a high-risk "all-or-nothing" academic environment.

- The Well-Rested Outlier: Interestingly, Well Rested students in this borderline category performed the worst (60.25). Within this specific "Survival Zone" context, a high amount of sleep may correlate with academic disengagement or a lack of the "emergency effort" seen in their poorly rested peers. However, the very small sample size (n=5) suggests these results should be viewed as an area for further longitudinal study rather than a definitive trend.
- Conclusion: For students at the C/D threshold, sleep is not a simple performance enhancer but a trade-off. The data highlights a punishing cycle: academic gains are currently being made at the cost of extreme anxiety and performance volatility. Institutional support should focus on helping these students achieve “Moderately Rested” status, which provides the best balance of academic consistency and emotional stability.

RECOMMENDATIONS

- Allocate resources to the “C/D” Borderline (1.9-2.1 GPA) students. Prioritize immediate counseling for the "Aspiring D" cohort. Data indicates this group faces a sharp 18.5% spike in anxiety compared to "Safe" students, suggesting the passing threshold is one of the primary psychological barriers to their academic performance.
- Expand tutoring efforts.With tutored students showing a 20.5% higher average score and a 25.9% reduction in grade volatility, tutoring acts as both an academic boost and an emotional stabilizer that ensures more consistent performance.
- Optimize for "Sustainable Effort" in Sleep: Encourage "Moderately Rested" habits to break the sleep deprivation/cramming cycle found in at-risk groups. Shifting away from poor rest reduces extreme performance volatility (Sigma 20.72) and helps lower peak anxiety levels, creating a more sustainable and predictable path to passing.

RECLECTION

While this analysis provides a robust snapshot of academic and emotional stressors, it also opens several avenues for deeper investigation into the student experience:

- Socioeconomic Factors & The "Sleep Paradox": I would like to integrate data regarding family income and part-time employment. I suspect these factors are the "missing variables" behind our sleep data; students in the "Survival Grind" may not be sacrificing rest by choice, but out of financial necessity. Investigating this correlation would provide a more holistic view of how external labor impacts internal academic performance.

- Grade Sensitivity Analysis: I am interested in mapping Final Exam scores against the Letter Grade (F–A) distribution. Analyzing the "weight" of these exams would reveal which students are most vulnerable to "High-Stakes Testing Anxiety." Understanding where a single exam score can pivot a letter grade would allow for even more precise identification of "Borderline" students.